PHYS 365 Final Exam Formula Sheet

OSCILLATIONS

$x(t) = x_m \cos(\omega t + \phi)$	$\omega = 2\pi f = \frac{2\pi}{T}$	$\omega = \sqrt{\frac{k}{m}}$	$\omega = \sqrt{\frac{g}{\ell}}$	$T = 2\pi\sqrt{\frac{\ell}{g}}$

WAVES

$\frac{\partial^2 y}{\partial^2 t} = v^2 \frac{\partial^2 y}{\partial x^2}$	$y(x,t) = f(x \pm vt)$	$y(x,t) = y_m \cos(kx \pm \omega t + \phi)$	
$v = \frac{\omega}{k} = f\lambda$	$k = \frac{2\pi}{\lambda}$	$v = \frac{c}{n}$	$\lambda = \frac{\lambda_0}{n}$
$v = \sqrt{\frac{\tau}{\mu}}$	$v = \sqrt{\frac{\gamma RT}{M}}$	$v = \sqrt{\frac{Y}{\rho}}$	$v = \sqrt{\frac{B}{\rho}}$
$Z = \sqrt{\mu\,\tau} = \frac{\tau}{v} = \mu v$	$Z = \sqrt{\rho B} = \rho v$	$P_{avg} = \frac{1}{2}\mu v \omega^2 y_m^2$	$\Delta p_m = (v\rho\omega)s_m$
$\Delta p = \Delta p_m \sin(kx - \omega t)$	$I = \frac{1}{2}\rho v \omega^2 s_m^2$	$I = \frac{P}{4\pi R^2}$	$I = \frac{P}{A}$
$\beta = (10\ dB)\log_{10}\left(\frac{I}{I_0}\right)$	$\Delta\beta = (10\ dB)\log_{10}\left(\frac{I_2}{I_1}\right)$	$I(r_2) = \left(\frac{r_1}{r_2}\right)^{n-1} I(r_1)e^{-\alpha\Delta r}$	
$R = \frac{Z_1 - Z_2}{Z_1 + Z_2}$	$T = \frac{2Z_1}{Z_1 + Z_2}$	$B = -\frac{\Delta p}{\left(\frac{\Delta V}{V}\right)}$	$Y = \frac{\left(\frac{F}{A}\right)}{\left(\frac{\Delta L}{L}\right)}$
$R_e = R^2$	$T_e = \frac{Z_2}{Z_1}T^2$	$\Delta p = -B\frac{\partial s}{\partial x}$	

WAVE INTERFERENCE

$y'(x,t) = \left[2y_m \cos\left(\frac{1}{2}\phi\right)\right]\sin\left(kx - \omega t + \frac{1}{2}\phi\right)$		$y'(x,t) = [2y_m \sin(kx)]\cos(\omega t)$
$\Delta\phi = \frac{\Delta L}{\lambda}2\pi$	$f_{beat} = f_1 - f_2$	$f' = f\frac{v \pm v_D}{v \pm v_S}$
Constructive interference $\frac{\Delta L}{\lambda} = 0,\ 1,\ 2, \ldots$		Destructive interference $\frac{\Delta L}{\lambda} = 0.5,\ 1.5,\ 2.5, \ldots$
Taut string or pipe open at both ends $f_n = \frac{v}{\lambda_n} = \frac{nv}{2L}$ for $n = 1,\ 2,\ 3, \ldots$		Pipe one end closed, one end open $f_n = \frac{v}{\lambda_n} = \frac{nv}{4L}$ for $n = 1,\ 3,\ 5, \ldots$
$d\sin(\theta) = m\lambda$ for $m = 0, 1, 2, \ldots$ (maxima-bright fringes)		
$d\sin(\theta) = \left(m + \frac{1}{2}\right)\lambda$ for $m = 0, 1, 2, \ldots$ (minima-dark fringes)		

$2t = m\frac{\lambda_0}{n}, \quad m = 1, 2, 3 \ldots$		$2t = \left(m - \frac{1}{2}\right)\frac{\lambda_0}{n}, \quad m = 1, 2, 3 \ldots$	
Reflection	Reflection phase shift	$a\sin(\theta) = m\lambda$ for $m = 1, 2, 3, \ldots$ (minima)	
Off lower index	0		
Off higher index	0.5 wavelength		
$I = 4I_0\cos^2\left(\frac{1}{2}\phi\right)$	$\phi = \frac{2\pi d}{\lambda}\sin(\theta)$	$\theta_R = \sin^{-1}\left(\frac{1.22\lambda}{d}\right)$	

MODERN PHYSICS

$F_e = \frac{1}{4\pi\epsilon_0}\frac{e^2}{r^2}$	$U = \frac{1}{4\pi\epsilon_0}\frac{e^2}{r}$	$U = qV$	$K = \frac{1}{2}mv^2$
$E = hf$	$p = \frac{h}{\lambda}$	$hf = K_{max} + \Phi$	$a_c = \frac{v^2}{r}$
$\Delta\lambda = \frac{h}{mc}(1 - \cos\phi)$	$\frac{1}{\lambda} = R\left(\frac{1}{n_{low}^2} - \frac{1}{n_{high}^2}\right)$	$K_{max} = eV_{stop}$	$R = \frac{P}{hf}$
$\Delta x\Delta p_x \geq \frac{h}{2\pi}$	$a_0 = \frac{\epsilon_0 h^2}{\pi m_e e^2}$	$E_n = -\frac{m_e e^4}{8\epsilon_0^2 h^2 n^2}$	$E_n = \frac{-13.6\text{ eV}}{n^2}$
$E_n = \frac{h^2}{8m_e L^2}n^2$	$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi}{L}x\right)$	$P(x_1, x_2) = \int_{x_1}^{x_2}\lvert\psi(x)\rvert^2 dx$	
$P(r) = \frac{4}{a_0^3}r^2 e^{\frac{-2r}{a_0}}$	$\psi(r) = \frac{1}{\sqrt{\pi}a_0^{3/2}}e^{\frac{-r}{a_0}}$	$T \approx e^{-2bL}$ $b = \sqrt{8\pi^2 m_e(U_b - E)/h^2}$	
$\frac{d^2\psi}{dx^2} + k^2\psi = 0$, with $k = \frac{2\pi\sqrt{2m_e(E-U)}}{h}$		$\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m_e}{h^2}\big(E - U(x)\big)\psi = 0$	

CONSTANTS

$c = 3.00 \times 10^8\ \frac{\text{m}}{\text{s}}$	$R = 8.314\ \frac{\text{J}}{\text{mol}\cdot\text{K}}$	$M_{air} = 28.8\ \frac{\text{g}}{\text{mol}}$	$g = 9.81\ \frac{\text{m}}{\text{s}^2}$
$v_{sound\ at\ 20°\text{C}} = 343\ \frac{\text{m}}{\text{s}}$	$T_K = T_{°\text{C}} + 273.15$	$\gamma_{monoatomic} = \frac{5}{3}$	
$I_0 = 1 \times 10^{-12}\ \frac{\text{W}}{\text{m}^2}$	$1\text{ atm} = 1.01 \times 10^5\text{ Pa}$	$\gamma_{diaatomic} = \frac{7}{5}$	
$m_e = 9.109 \cdot 10^{-31}\text{ kg}$		$m_p = 1.67 \cdot 10^{-27}\text{ kg}$	
$a_0 = 0.5292 \cdot 10^{-10}\text{ m}$		$1\ u = 1.66054 \cdot 10^{-27}\text{ kg}$	

$h = 6.63 \cdot 10^{-34}\ \mathrm{J \cdot s} = 4.136 \cdot 10^{-15}\ \mathrm{eV \cdot s}$	$\epsilon_0 = 8.854 \cdot 10^{-12}\ \frac{\mathrm{C^2}}{\mathrm{N \cdot m^2}}$
$\sigma = 5.67 \cdot 10^{-8}\ \frac{\mathrm{W}}{\mathrm{m^2 \cdot K^2}}$	$e = 1.602 \cdot 10^{-19}\ \mathrm{C}$
$k_B = 1.381 \cdot 10^{-23}\ \frac{\mathrm{J}}{\mathrm{K}} = 8.617 \cdot 10^{-5}\ \frac{\mathrm{eV}}{\mathrm{K}}$	Rydberg constant $R = 1.09737 \cdot 10^{7}\ \mathrm{m^{-1}}$
MATHEMATICAL FORMULAE	
$\cos(\alpha) \pm \cos(\beta) = 2\cos\left(\frac{1}{2}(\alpha \pm \beta)\right)\cos\left(\frac{1}{2}(\alpha \mp \beta)\right)$	
$\sin(\alpha) \pm \sin(\beta) = 2\sin\left(\frac{\alpha \pm \beta}{2}\right)\cos\left(\frac{\alpha \mp \beta}{2}\right)$	
$\cos(\alpha \pm \beta) = \cos(\alpha)\cos(\beta) \mp \sin(\alpha)\sin(\beta)$	$\tilde{A} = Ae^{j\theta} = A(\cos(\theta) + j\sin(\theta))$
$\sin(\alpha \pm \beta) = \sin(\alpha)\cos(\beta) \pm \cos(\alpha)\sin(\beta)$	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
$\sin(-\alpha) = -\sin(\alpha)$	$\cos\left(\alpha \pm \frac{\pi}{2}\right) = \mp\sin(\alpha)$
$\cos(-\alpha) = \cos(\alpha)$	$\sin\left(\alpha \pm \frac{\pi}{2}\right) = \pm\cos(\alpha)$
$\int \cos(ax) = \frac{\sin(ax)}{a} + c$	$\int \sin(ax) = -\frac{\cos(ax)}{a} + c$
$\int \sin^2(ax)\,dx = \frac{x}{2} - \frac{\sin(2ax)}{4a} + c$	$\int \cos^2(ax)\,dx = \frac{x}{2} + \frac{\sin(2ax)}{4a} + c$
$\sin^2\theta = \frac{1}{2}(1 - \cos(2\theta))$	$\cos^2\theta = \frac{1}{2}(1 + \cos(2\theta))$